# Token cost of reading equations: PDF page images versus Markdown transcription, and its share of a Pro 5-hour window

Multipoint Correlation Functions project — working note

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## Question and set-up

How many input tokens does it cost Claude to read a displayed equation from a PDF (as an image, following the project's PDF-reading protocol) compared with reading the same equation from a `pdf-to-markdown` transcription (as LaTeX text)?

**Same content on both sides.** The test document is Lihm *et al.*, PRB **109**, 125138 (2024), whose transcription (`_J__M__Lihm_2024__....md`) contains 274 display blocks (270 tagged equations). The original PDF is no longer mounted, so every display block of the `.md` was typeset with `pdflatex` at PRB column geometry (3.4 in column, 10 pt Times via `mathptmx`), rendered at 200 dpi, and cropped to full column width plus 8 px padding. *Check of realism:* the real display equations of the mounted Kugler 2021 and Lee 2021 PDFs, located by their tags in the text layer, have median crop heights of 90 px and 74 px at 200 dpi; the typeset median is 74 px.

**Counting rules.** Image tokens follow Anthropic's published rule

$$N_{\text{img}} = \left\lceil \frac{w}{28} \right\rceil \left\lceil \frac{h}{28} \right\rceil , \tag{1}$$

with $w,h$ in pixels, on the high-resolution tier (Claude 4.7 and later): long edge $\leq 2576$ px and $N_{\text{img}} \leq 4784$, larger images being downscaled. Text tokens were counted with the legacy Claude tokenizer shipped in the `anthropic` Python SDK 0.34.2, because no API key was available for the token-counting endpoint. Current models use a newer tokenizer that yields roughly 1.0–1.35× as many tokens for the same text (Anthropic), with independent measurements up to $\approx 1.47\times$ on technical text; so the Markdown figures below should be multiplied by 1.0–1.5. All numbers are computed, not metered from a live context.

## Results

| what is read (per display block) | p10 | median | p90 | sum (274) |
|---|---|---|---|---|
| `.md`: LaTeX block alone | 33 | 64 | 162 | 23.7k |
| `.md`: grep hit with ±3 lines context | 126 | 206 | 375 | — |
| PDF crop, native 200 dpi | 50 | 75 | 175 | 26.4k |
| PDF crop, 2.5× upscale | 305 | 427 | 1037 | 154k |
| PDF crop, 3× upscale (protocol default) | 438 | 614 | 1460 | 220k |
| PDF crop, 4× upscale | 644 | 920 | 2208 | 338k |
| full page view (1700 × 2200 px) | | 4758 | | |

Table 1: Tokens per display equation. Text tokens use the legacy tokenizer (scale by 1.0–1.5); image tokens use Eq. (1).

**Upscaling sets the price.** At native resolution a crop costs about as much as its LaTeX (median ratio $N_{\text{img}}/N_{\text{text}} = 1.16$). Image tokens grow as the square of the scale factor $s$, $N_{\text{img}} \propto s^2$, and the protocol upscales 2.5–4× so that indices, bars and primes are legible. At $s = 3$ the median ratio is 9.5 (p10 7.2, p90 13.3).

**Example: Eq. (12a).** From the transcription (line 173):

```
i\omega G[\mathcal{O}^1,\mathcal{O}^2](i\omega)=G[[\mathcal{O}^1,H],
\mathcal{O}^2](i\omega)+\langle[\mathcal{O}^1,\mathcal{O}^2]_{\zeta_1^2}\rangle, \tag{12a}
```

costs 71 tokens (185 with ±3 lines of context). The 3× crop of the same equation (2040 × 249 px) costs 657 tokens:

$$i\omega G[\mathscr{O}^1,\mathscr{O}^2](i\omega) = G[[\mathscr{O}^1,H],\mathscr{O}^2](i\omega) + \langle[\mathscr{O}^1,\mathscr{O}^2]_{\zeta_1^2}\rangle, \tag{12a}$$

**Realistic workflows per equation.**

| workflow | tokens (median) |
|---|---|
| transcription: grep the tag, print the block with context | ≈ 200 |
| PDF, best case: text-layer locate + one 3× crop | ≈ 800 |
| PDF, typical: locate + full-page view to choose the box + 3× crop | ≈ 5600 |

That is a factor of about 4 (best case) to 27 (typical) in favour of the transcription. Long equations widen the gap: the largest block, Eq. (100), costs 532 text tokens versus 3431 image tokens.

**Whole paper.** The entire `.md` (prose included) is 64.6k tokens. Reading all 274 equations as 3× crops is 220k; viewing all 32 pages at native resolution is 152k, and that resolution is too coarse to settle subscripts.

## Share of a Pro 5-hour window

**Budget.** Anthropic publishes no token size for the Pro 5-hour session limit. The best public estimate found comes from one user's Claude Code logs (FinOps LLM): the Max 5x session limit stopped at a median of about $37 of API-equivalent usage in September 2026, which projects to about $B \approx \$7.4$ for Pro (Max 5x is 5× Pro per session). This is a projection, not a Pro measurement. The Opus 5.5 launch (22 September 2026) raised five-hour limits by an unpublished amount, so $B$ is likely a lower bound. The estimate also assumes that Anthropic's usage units scale like API dollars.

**Prices and context mechanics.** Opus 5.5 list prices per million tokens: input $4, 5-minute cache write $5, cache read $0.20, output (including thinking) $20. In chat, every tool call and every turn re-sends the whole context. New content is charged once (taken here at the $5 write price) and is then re-read at $0.20 per million on every later step for as long as the conversation lasts. A question therefore costs approximately

$$C \approx 5\times10^{-6} N_{\text{new}} + 0.2\times10^{-6} \sum_{\text{steps}} N_{\text{ctx}} + 20\times10^{-6} N_{\text{out}} \quad (\$), \tag{2}$$

where $N_{\text{new}}$ is the new content entering the context, $N_{\text{ctx}}$ is the context size at each step, and $N_{\text{out}}$ is output plus thinking.

| read | tokens | first-read cost | share of $B$ | reads per window |
|---|---|---|---|---|
| `.md`: tag grep with context | 210–310 | $0.0010–0.0015 | ≈ 0.02% | ∼5000–7000 |
| PDF best case (locate + 3× crop) | 810–910 | ≈$0.0043 | ≈ 0.06% | ∼1700 |
| PDF typical (+ full-page view) | ≈5600 | ≈$0.028 | ≈ 0.4% | ∼260 |
| whole `.md` of Lihm 2024 | 65k–97k | $0.32–0.48 | 4–7% | — |
| all 32 pages as images | 152k | $0.76 | ≈ 10% | — |
| all 274 equations as 3× crops | 220k | $1.10 | ≈ 15% | — |

Table 2: First-read cost against $B = \$7.4$. Text tokens include the 1.0–1.5 tokenizer factor. Whole-paper image reading is impractical anyway: claude.ai accepts 20 images per message.

**Re-reads often exceed the first read.** A typical PDF equation read (5.6k tokens) adds about 0.015% of $B$ per later step. The whole transcription (≈97k tokens) adds about 0.26% per step, so about 13% over 50 later steps, which is more than loading it cost. Page images left in context compound the image route's disadvantage, and grepping single equations beats opening a whole transcription.

## Check against observed usage

The observed consumption in this project is 1–5% of the Pro window per question that does not create a file. Equation (2) reproduces this once the fixed context is included. That context comprises the system prompt with tool definitions, the project instructions, the memory listing and the conversation history; it is not measurable from inside the chat and is taken as about 40–50k tokens.

| case | share of $B$ |
|---|---|
| light question, warm cache: 1 step, 50k re-read, 2k output | $\approx 0.7\%$ |
| medium question: 5 tool calls adding 25k, context 50k$\rightarrow$75k, 3k output | $\approx 3$–$4\%$ |
| same question after the cache expired: 50k prefix rewritten | $\approx 6$–$7\%$ |

The predicted range of about 1–7% brackets the observed 1–5%. Within a question the cost is set mainly by the number of tool steps, the fixed context and the cache state. Equation reading adds a small amount: about 0.02% from the transcription, about 0.4% for a typical PDF read. Whether claude.ai caches for 5 minutes or 1 hour is unknown, and this decides whether pausing between questions is expensive. Because the fixed context and the cache state are unknown, the check confirms $B$ only to within a factor of about 2. Reading the usage meter before and after three controlled questions would isolate each term: one with no tool calls, one reading a single equation from the `.md`, and one viewing a PDF page.

## Caveats

The transcription is cheap because it is trusted as source; the image-reading cost of producing it was paid once at conversion and is not counted here. Whether a cheaper image route, such as rendering at 300 dpi without upscaling (about 2.25$\times$ the 200 dpi crop instead of 9$\times$), reads indices reliably was not tested. The Pro budget $B$ is an indirect estimate. The assumption that usage units scale like API dollars is unverified.

## Sources


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